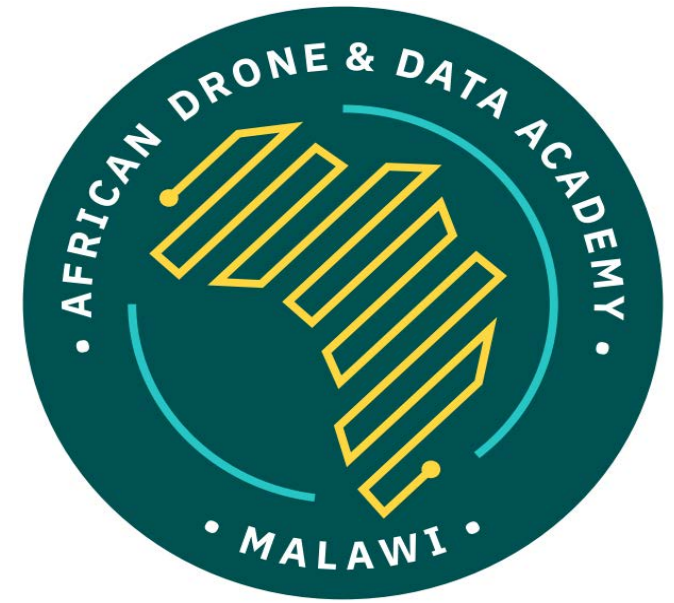




TOP Level 1

Day 7 Version A





TOP Level 1

Standard Operating Procedures Part 2



Pre-flight checklists - flight planning

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- As part of the feasibility assessment for the flight, the operator should review the operational airspace and determine that the flight is authorized
 - Airspace rules require that the drone stay outside certain areas in order to be compliant with operations that serve all aircraft, both manned and unmanned
 - Altitude restrictions and operations in the proximity of aerodromes are controlled by the civil aviation authority



Pre-flight checks - Air law: Check compliance with the applicable operational environment

- 7. **SOPs**
- 8. SMS
- 9. Work safety
- 10. Crew roles

- Operator limitations:

- 30.
 - 1) Notwithstanding the provisions of sub regulation (1), operations for private and commercial categories of RPA may be conducted at such heights and lateral distances as the Authority may approve.
 - 2) A person shall not operate an RPA unless approved by the Authority
 - a) In conditions other than Visual Meteorological Conditions (VMC)
 - b) At night; and
 - c) Beyond Visual Line of Sight (B-VLoS).

Pre-flight checks - Air law: Check compliance with the applicable operational environment

- 7. **SOPs**
- 8. SMS
- 9. Work safety
- 10. Crew roles

- Operations in controlled airspace:

- 32.
 - 1) No RPA may be operated in controlled airspace, except by the holder of an ROC and as approved by the Authority
 - 2) The Authority may approve an RPA operation in controlled airspace as contemplated in Aviation (Rules of the Air and Air Traffic Control) Regulations currently in force, only in
 - a) VMC in an ATZ and CTR below 400feet; and
 - b) Subject to compliance with the conditions prescribed in the Malawi Aviation Regulations currently in force.

Pre-flight checks - Air law: Check compliance with the applicable operational environment

- 7. **SOPs**
- 8. SMS
- 9. Work safety
- 10. Crew roles

- Operations in controlled airspace:

32. Continued

3. RPA operating in controlled airspace shall have the same ability to automatically detect-and-avoid traffic/ obstacles, as other aircraft while moving through the air.
4. ATC communication:
 - a) RPA pilots shall ensure that ATC is made aware of any operations that shall take place in areas which are likely to affect manned and controlled air traffic;
 - b) The ATC shall establish procedures, acceptable to the Authority, for integration of RPA operation into the airspace to ensure aviation safety. Such procedures shall include communication and surveillance detection; and
 - c) Procedures referred to in sub-regulation (b) shall prescribe required information to be passed to ATC by RPA pilot before and during RPA operations.

Pre-flight checks - Air law: Check compliance with the applicable operational environment

- 7. SOPs
- 8. SMS
- 9. Work safety
- 10. Crew roles

- Operations in the vicinity of an aerodrome:

33. A person shall not operate an RPA:

- a) within 10 kilometers of an aerodrome from the aerodrome reference point for code C, D ,E and F aerodromes;
- b) within 7 kilometers of an aerodrome from the aerodrome reference point for code A and B aerodromes;
- c) on approach and take-off paths;
- d) within the vicinity of navigation aids;
- e) within the aerodrome traffic zone;
- f) within terminal traffic holding patterns;

Pre-flight checks - Weather analysis: **forecast the day before and observations the day of flight**

7. SOPs
8. SMS
9. Work safety
10. Crew roles

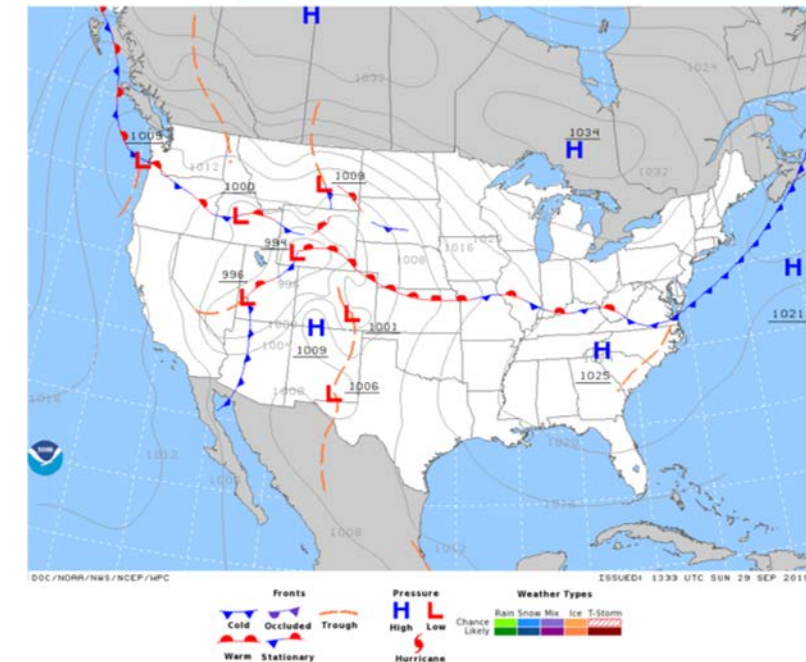
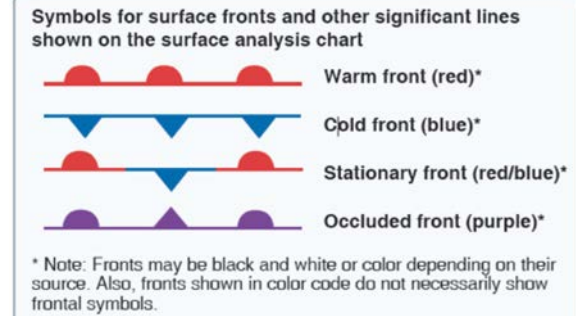
- Weather briefings provide both a chance to check compliance with air law (minimum flight visibility, for instance) and the likelihood that the flight will be feasible (wind speed that exceeds the aircraft operational limits)
- Weather information is either obtained from government sources or commercial services that provide details about surface conditions (on the ground) and synoptic conditions (atmospheric trends) useful for flight planning



A weather forecast is a critical part of the day-before flight planning

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Forecasts are based on large-scale weather models, shown on surface analysis charts that depicts fronts, high pressure systems and low pressure systems
- These systems define the behavior of winds and the trends that can be expected during the day that will affect flying conditions
- Generally speaking:
 - Fronts bring cloudy conditions
 - High pressure brings favorable conditions
 - Low pressure brings unfavorable conditions



A weather forecast is a critical part of the day-before flight planning

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Forecasts are also known as **TAFs** (terminal aerodrome forecast)
- They can be obtained by the South African Weather Service
- The TAF is usually accompanied by an observation report called the **METAR** (meteorological aviation routine report)
- Both the METAR and TAF use a coded system for describing weather conditions

```
FWKI - Lilongwe International
METAR
FWKI 291600Z 06008KT CAVOK 25/11 Q1016
NOSIG
TAF
FWKI 291500Z 2918/3024 09012KT CAVOK
BECMG 3004/3006 10006KT 8000 HZ SKC
BECMG 3010/3012 08008KT 9999 FEW030
FM301400 08008KT CAVOK
```

The codes used in METARs and TAFs should be understood since they contain important weather information

7. SOPs
8. SMS
9. Work safety
10. Crew roles



Key to Aerodrome Forecast (TAF) and Aviation Routine Weather Report (METAR) (Back)



	In METAR , ReMarK indicator & remarks. For example: <u>Sea- Level Pressure</u> in hectoPascals & tenths, as shown: 1004.5 hPa; <u>Temp/dew-point</u> in tenths °C, as shown: temp. 18.2°C, dew-point 15.9°C	RMK SLP045 T01820159
FM091930	<u>From</u> : changes are expected at: 2-digit date, 2-digit hour, and 2-digit minute beginning time: indicates significant change. Each FM starts on a new line, indented 5 spaces	
TEMPO 0920/0922	<u>TEMPO</u> rary: changes expected for <1 hour and in total, < half of the period between the 2-digit date and 2-digit hour beginning, and 2-digit date and 2-digit hour ending time	
PROB30 1004/1007	<u>PROB</u> ability and 2-digit percent (30 or 40): probable condition in the period between the 2-digit date & 2-digit hour beginning time, and the 2-digit date and 2-digit hour ending time	
BECMG 1013/1015	<u>BEC</u> o <u>MinG</u> : change expected in the period between the 2-digit date and 2-digit hour beginning time, and the 2-digit date and 2-digit hour ending time	

The codes used in METARs and TAFs should be understood since they contain important weather information

7. SOPs
8. SMS
9. Work safety
10. Crew roles

Table of Significant Present, Forecast and Recent Weather - Grouped in categories and used in the order listed below; or as needed in TAF, <u>No Significant Weather</u> .			
Qualifiers			
Intensity or Proximity			
“-” = Light		No sign = Moderate	“+” = Heavy
“VC” = Vicinity, but not at aerodrome. In the US METAR, 5 to 10 SM from the point of observation. In the US TAF, 5 to 10 SM from the center of the runway complex. Elsewhere, within 8000m.			
Descriptor			
BC – Patches	BL – Blowing	DR – Drifting	FZ – Freezing
MI – Shallow	PR – Partial	SH – Showers	TS – Thunderstorm
Weather Phenomena			
Precipitation			
DZ – Drizzle	GR – Hail	GS – Small Hail/Snow Pellets	
IC – Ice Crystals	PL – Ice Pellets	RA – Rain	SG – Snow Grains
SN – Snow	UP – Unknown Precipitation in automated observations		
Obscuration			
BR – Mist ($\geq 5/8$ SM)	DU – Widespread Dust	FG – Fog ($< 5/8$ SM)	FU – Smoke
HZ – Haze	PY – Spray	SA – Sand	VA – Volcanic Ash
Other			
DS – Dust Storm	FC – Funnel Cloud	+FC – Tornado or Waterspout	
PO – Well developed dust or sand whirls		SQ – Squall	SS – Sandstorm
- Explanations in parentheses “()” indicate different worldwide practices. - Ceiling is not specified; defined as the lowest broken or overcast layer, or the vertical visibility. - NWS TAFs exclude BECMG groups and temperature forecasts, NWS TAFS do not use PROB in the first 9 hours of a TAF; NWS METARs exclude trend forecasts. US Military TAFs include Turbulence and Icing groups.			

Generally speaking, the TAF is used for day before flight planning, and both the TAF and METAR are used the day of the flight

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Winds, visibility and ceiling are the most important information for flight planning
 - Most countries only allow RPAs to fly up to 400 ft. AGL (above ground level), so clouds are usually not a problem
 - The operator must always maintain visual line of sight of the aircraft, meaning that if the visibility is poor, then the flight radius will be small
 - Winds are reported as steady and gusts (02010KTG20). The maximum predicted windspeed (steady or gust) cannot exceed the operational limits of the aircraft



Trial METAR interpretation

7. SOPs
8. SMS
9. Work safety
10. Crew roles



AVIATION WEATHER CENTER

NOAA NATIONAL WEATHER SERVICE

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METAR Data

[METAR Home](#) [Plot](#) [Data](#) [Board](#) [Info](#)

IDs: **Format:** ☐ Raw ☒ Decoded [most recent only](#) ☐ Include TAF [Print](#) [Update](#)

Data at: 2117 UTC 29 Sep 2019

METAR for: KBCB (Blacksburg/VA Tech A, VA, US)

Text: **KBCB 292115Z AUTO 12007KT 10SM SCT065 BKN080 BKN100 29/19 A3025 RMK AO2 T02940190**

Temperature: 29.4°C (85°F)

Dewpoint: 19.0°C (66°F) [RH = 54%]

Pressure (altimeter): 30.25 inches Hg (1024.5 mb)

Winds: from the ESE (120 degrees) at 8 MPH (7 knots; 3.6 m/s)

Visibility: 10 or more sm (16+ km)

Ceiling: 8000 feet AGL

Clouds: scattered clouds at 6500 feet AGL, broken clouds at 8000 feet AGL, broken clouds at 10000 feet AGL

QC Flag: automated observation with no human augmentation

The aircraft condition inspection occurs the **day before** and also on the **day of** the flight

- 7. SOPs
- 8. SMS
- 9. Work safety
- 10. Crew roles

- The aircraft condition inspection ensures the integrity of the aircraft and its ability to perform a flight mission.

Multirotor Checks:

- a. Preflight Hardware checks
 - i. Prop condition good? Yes / No
 - ii. Prop bolts secure? Yes / No
 - iii. Motor bolts secure? Yes / No
 - iv. Frame integrity check? Yes / No
 - v. Battery mounts secure? Yes / No
 - vi. Balance aircraft on roll and pitch axes – ensure CG is correct Yes / No
- b. Power-up checks (batteries may not be fully charged for this check)
 - i. No errors in power-up. Notes: _____
 - ii. Roll and pitch the aircraft – check attitude in QGroundControl. Check []
 - iii. Check motor sequence using built-in test. Correct? Yes / No

Aircraft condition inspection (pg. 2)

- 7. SOPs
- 8. SMS
- 9. Work safety
- 10. Crew roles

c. All Up Weight _____ [Circle as applicable: without Batteries without Payload]

d. Batteries (visual inspection and put on charge):

i. Aircraft

Yes / No / NA

Number of pairs of batteries needed for planned flights: _____

Cells:

Capacity:

Quantity:

ii. Transmitter

Yes / No / NA

Payload: _____

a. Securely mounted?

Yes / No

b. Power?

Yes / No

c. Functional check?

Yes / No

d. Connectors connected and secure?

Yes / No / NA

e. Internal components secure?

Yes / No / NA

f. Payload weight or status _____

The **day of** flight planning process is similar to the **day before** pre-flight check

- 7. **SOPs**
- 8. SMS
- 9. Work safety
- 10. Crew roles

Multicopter Checks:

Hardware Checks

Yes / No

- a. Prop condition good?
- b. Props installed in correct orientations?
- c. Motor bolts secure?
- d. Motors rotate smoothly?
- e. Frame integrity check?
- f. CG check (balance in roll and pitch)
- g. Batteries strapped in?
- h. Batteries (visual check and charged?):
 - i. Aircraft

Yes / No

Yes / No

Yes / No

Yes / No

Yes / No

Check []

Yes / No

Yes / No

Preflight Voltage (Cells):

Capacity:

Quantity:

- i. Transmitter

Yes / No

Day of aircraft checks continued

- 7. SOPs
- 8. SMS
- 9. Work safety
- 10. Crew roles

Software Checks

Yes / No

- i. Radio failsafe check?
- j. Telemetry operational?

Yes / No / NA

Yes / No / NA

3. Payload Name/Description: _____

- a. Securely mounted?
- b. Power?
- c. Function check?
- d. Connectors secure?
- e. Internal components secure?
- f. Weight? _____

Yes / No

Yes / No

Yes / No

Yes / No / NA

Yes / No / NA

Before takeoff, the remote pilot in command (RPIC) conducts a briefing with the flight crew

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- This briefing shares important information about the flight with the flight crew
 - The briefing is outlined in form AC01 and is guided by *ADDAttracking*
 - A document check is completed, the flight is categorized, and responsibilities are assigned to the crew
 - All flight planning and checklists are verified to be complete
 - Safe operational flight distances are reviewed



Aircraft	Obstacle	Safe distance
< 30 kts top speed	People (non-participating)	30m, no overflight
	People (participating) or non-participating inside structures	2m (< 0.5 kg aircraft) 4m (>1 kg aircraft)
	Vehicles, structures and trees	30m
> 30 kts top speed	People (non-participating)	30m, no overflight
	People (participating) or non-participating inside structures	10m
	Vehicles, structures and trees	30m

Before takeoff, the remote pilot in command (RPIC) conducts a briefing with the flight crew

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- For students receiving instruction, it is during this briefing that the remote pilot instructor (RPI) will explain the maneuvers that will be practiced
 - A lesson plan will be presented which has specific objectives that must be completed for a successful lesson
- If a visual observer (VO) is assigned, then during this briefing the communications protocols will be agreed upon to handle routine and emergency situations



Before takeoff, the remote pilot in command (RPIC) conducts a briefing with the flight crew

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- The Airmanship Principles are reviewed with the flight crew:
 - Know your aircraft, the capabilities and limitations in normal and non-normal scenarios.
 - Know yourself, your limitations and personal habits that effect safety.
 - Know your environment, the airspace, the weather, the operation.
 - Know your team, conduct briefings, use your observers, communicate, be positive and resolve conflicts quickly and fairly.
 - Know the Risks, always conduct a thorough risk assessment.
 - Stay Proficient, practice and conduct 'dry runs' of complex operations, and scenario based training for emergencies.
 - Have the discipline to keep up to date on the latest safety and operational knowledge, including technical information regarding your platform.

Non-normal scenarios and emergency procedures are just as important to the mission as are the routine operations

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Whenever there is a deviation in flight operations that is not anticipated, the pilot should have a pre-memorized procedure to follow since there will not be any time to refer to the aircraft SOPM or the user manual
- In general, the pilot should “fly the aircraft” first, and resolve the anomaly second
- A landing is to be executed as soon as practical, even if the anomaly is corrected in flight.
 - A repeat of the anomaly could happen, and the outcome may not be as good the second time as it was the first time

Non-normal scenarios and emergency procedures are just as important to the mission as are the routine operations

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- For the S-500, the SOPM has the following recommended action for emergencies:
 - Low Endurance
 - Immediate Landing in Controlled Area
 - Loss of GCS or Radio Communications
 - Immediate RTL at Communication Regain
 - Loss of Flight Control
 - Immediate Change to Stabilize and Landing (if possible).
 - Loss of Visual Line of Sight
 - Regain LOS and Navigate Back to Home



Non-normal scenarios and emergency procedures are just as important to the mission as are the routine operations

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- An Emergency Response Plan (ERP) is also available in the Safety Management System (SMS) Manual
- Emergency Procedures Quick Reference Checklist:
 - If necessary, call emergency services.
 - Render first aid as necessary.
 - Contact the ADDA Head of Training or the ADDA Project Manager via the call out list.
 - Preserve accident/incident site to ensure aircraft wreckage is not tampered with.
 - Secure the Ground Control Station.
 - Once the flight crew and scene are secure/safe take pictures and begin documentation of the incident.

Non-normal scenarios and emergency procedures are just as important to the mission as are the routine operations

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- An Emergency Response Coordinator (ERC) is always designated (by the Aviation Safety Officer) at the site to take the lead on responding to an emergency, accident or incident
 - This person has the responsibility to prepare for emergencies by providing the following at the flying site:
 - First aid kit that includes enough supplies to handle expected injuries from drone flight
 - A phone number of the nearest emergency response unit for transportation to a hospital
 - 2l of drinking water
 - Clean patient space to administer first aid
 - A bucket of sand for putting out fires

Non-normal scenarios and emergency procedures are just as important to the mission as are the routine operations

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- An Emergency Contact Tree is used to sequentially contact personnel affiliated with the ADDA to activate an emergency response:
 - First, contact the Emergency Response Coordinator (ERC) in the event of an incident or accident. This person is responsible for planning ahead for emergency situations
 - Next, the Head of Training is contacted
 - Next, contact the Project Manager if the Head of Training cannot be reached. The Project Manager also serves as the Aviation Safety Officer Both the Head of Training and the Project Manager should be on-site
 - Next, contact the Chief Pilot who may or may not be on-site
 - Next, contact the Programmatic Lead who may or may not be on-site

Non-normal scenarios and emergency procedures are just as important to the mission as are the routine operations

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- **Mandatory Incident/Accident Reporting System:** All UAS operators are required to submit a complete and detailed incident/accident report if any of the following occur during operations:
- ***Incidents***
 - Near miss with an object or person.
 - Crash with any amount of damage beyond normal wear and tear.
 - Reputation conflict.
- ***Accidents***
 - Property damage of any type/cost
 - Injury of any type.
- All reports received by the ADDA will be reviewed and addressed by members of the Safety Committee.

Non-normal scenarios and emergency procedures are just as important to the mission as are the routine operations

7. SOPs

8. SMS

9. Work safety

10. Crew roles

The incident/accident reporting form must be filled out immediately after the event:

The Incident/Accident involved (check all that apply):

☐	Injury	☐	Prop Strike/Tipover	☐	Loss of Data Downlink
☐	Death	☐	Foreign Object Damage	☐	Airspace Violations
☐	Property Damage	☐	Wildlife	☐	Procedure Error
☐	Traffic Conflict in Flight	☐	Near Midair Collision	☐	Human Factor Error
☐	Fatigue	☐	Hard Landing	☐	Other
☐	Fuel/Energy Event	☐	Loss of Command Uplink	☐	
☐	Altitude Deviations	☐	Loss of Telemetry Downlink	☐	

Classification (check all appropriate responses):

☐	Damage to Aircraft (salvageable)	☐	Injury to Person (no hospital)
☐	Damage to Aircraft (hull loss)	☐	Injury to Person (hospitalized)
☐	Lost Aircraft (unrecoverable)	☐	Death of Person
☐	Damage to Property (<\$500)		
☐	Damage to Property (>\$500)		

Post-flight and Pack-up

- 7. SOPs
- 8. SMS
- 9. Work safety
- 10. Crew roles

- Securing the aircraft
 - After a flight mission has ended, it is necessary to secure the aircraft and conduct a post-flight check of the aircraft
- Post-flight inspection
 - The purpose of a post-flight inspection is to find if anything changed on the aircraft during the flight which could affect the safety of the next flight
 - Abnormal sounds during flight may indicate a developing problem
 - Screws and wiring can loosen up during a flight
 - The props are always inspected after landing even though they will be inspected again before the next flight
 - Battery condition should be checked after landing to determine the state of charge

Post-flight and Pack-up

7. SOPs
8. SMS
9. Work safety
10. Crew roles

- Any noted anomalies in the post-flight inspection are entered via *ADDAttracking*
- The remote pilot in command will debrief the crew on the flight
 - The RPIC will note any interesting control or operational conditions that should be shared with the crew
 - Likewise, if the flight crew has anything to share, they will share it at this time
 - Logbooks are updated at this time by the RPIC (through *ADDAttracking*):

Logbook	For what?	Who maintains?
Maint_S500_SNXX	Maintenance logging of S-500 aircraft	RPI or designee
Flightlog_S500_SNXX	Aircraft logbook	RPIC
Bat_SNXX	Battery logbook	RPIC
Pilot <Name> Logbook	Pilot logbook	RPIC or SP